

COULD PLASTIC BE A FEASIBLE ENERGY SOURCE ?

ABSOLUTLY!!!

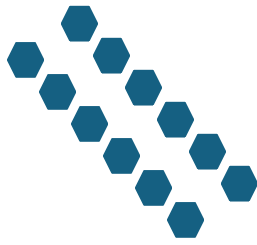
PRESENTING YOU

POLYCYCLE POWER



SCENARIO ANALYSIS

THE STATE OF ENERGY



4E ENERGY PROBLEM



ENOUGH

Energy is in **scarcity** due to macroeconomic circumstances



ENVIRONMENTAL

80% global energy consumption is via **fossil fuels**



ECONOMICAL

Rising prices of Natural gas, Electricity & Other Energy Sources



EQUITABLE

Energy is an **inaccessible commodity** to many

WE NEED **NEW ENERGY**
BUT WHICH ONE?

ALTERNATIVE
EVALUATION

Fossil Fuel

Solar Energy

Hydro Power

Wind Energy

Nuclear Energy

**Biomass
Energy***

Enough



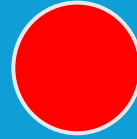
Equitable



Environmental



Economical



More
about
**Biomass
Energy**



Input: Organic,
Compostable,
Food Waste



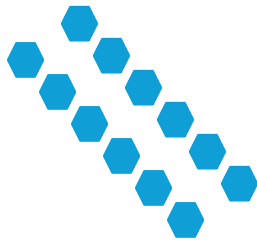
Potential Power
Generation
1500 MWe



Potential alt. of
LPG market
(14.41 lakh ton)



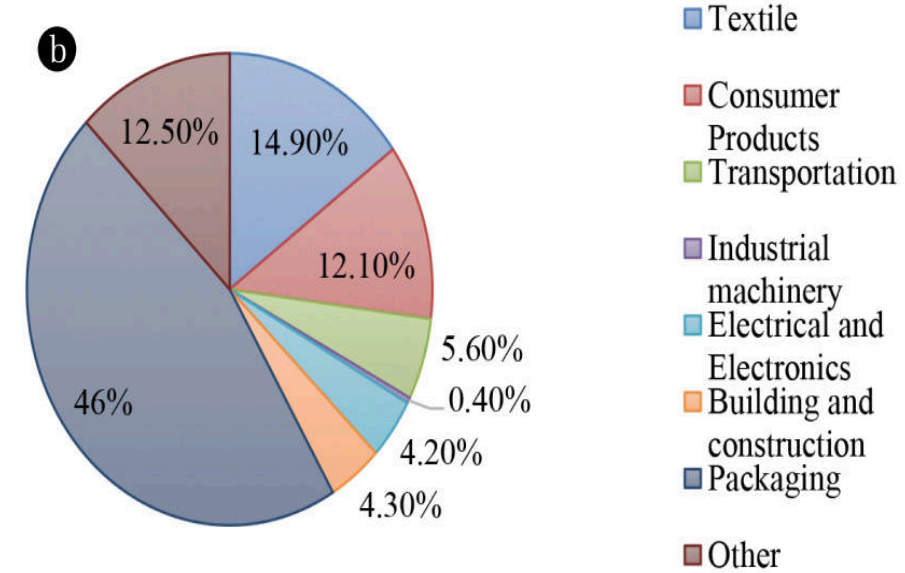
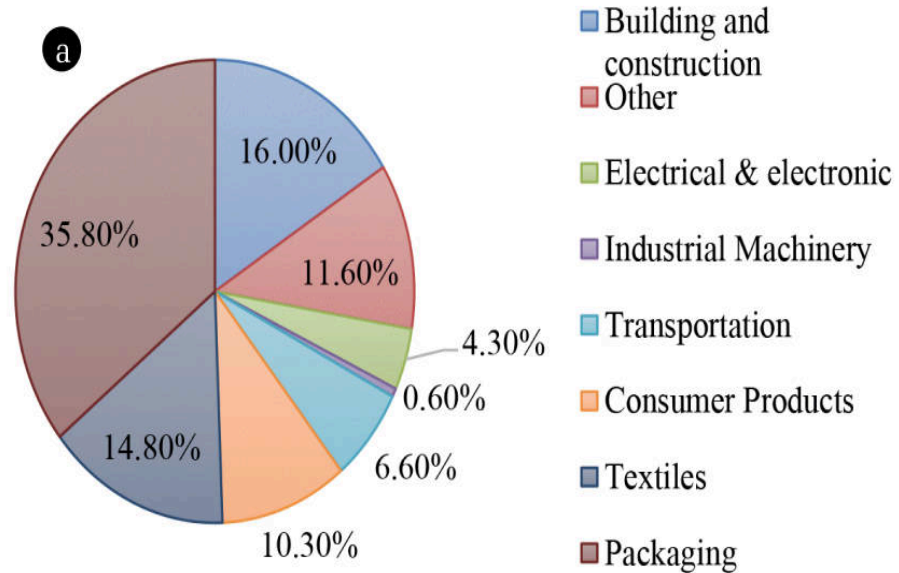
PROBLEM STATEMENT



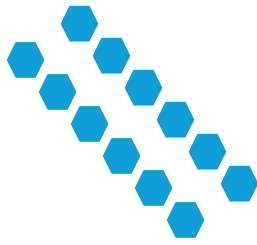
BIO-DEGRADABLE
PLASTIC

PLASTIC TO POWER

Packaged
Bio-Fuel

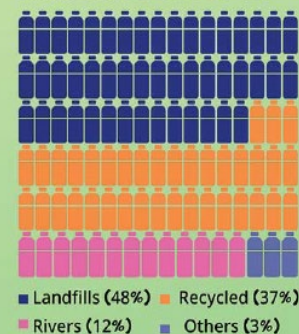


PROBLEM STATEMENT

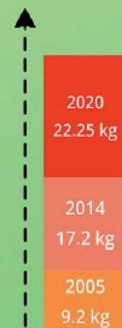


Overview of plastic pollution in Bangladesh

The total plastic waste in Dhaka was **646 tons** in 2020



Bangladesh was 10th among top 20 mismanaged plastic waste generating nations in 2010



Per capita plastic consumption has more than doubled in Dhaka in the last 15 years.



Effects of plastic pollution

Plastic articles degrade into microplastic, which endangers marine biodiversity significantly



Due to plastic pollution, one-third of Dhaka's canals cannot transport rainwater into rivers



In Bangladesh, plastic waste is often incinerated which leads to air pollution and human health hazard



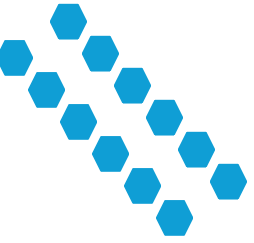
The estimated clean-up cost of plastic waste for the government was as high as 30% of the total revised budget for the MoEFCC in 2020



800,000 Tons of plastic waste is produced in Bangladesh each year
- [_statistica](#)

Less than half of the plastic waste in Bangladesh gets recycled while 48 per cent reaches landfills, and the rest is either dumped in rivers or discarded in drains and in other areas of the city corporations (The World Bank , 2021).

TARGET GROUP ANALYSIS



DEMOGRAPHIC



32-55



SEC C1
B2 B1 A3

GEOGRAPHIC



Urban
Suburban

VALUES & LIFESTYLE

BEHAVIORAL



Purchase Frequency
Monthly/ Bi-Monthly



Usage Frequency:
Daily/Frequently



Benefits Sought:
Effectiveness, Safety



Household Decision
Maker/Homemaker



Savings
Mindset



Seeks
Convenience



Cares for
Planet



BIG IDEA



INTRODUCING POLYCYCLE POWER

RENEWABLE BIO-GAS CYLINDERS

“POWERING PROGRESS WITH PLASTICS”

29%

Cost
Saving

80%

Less
Carbon



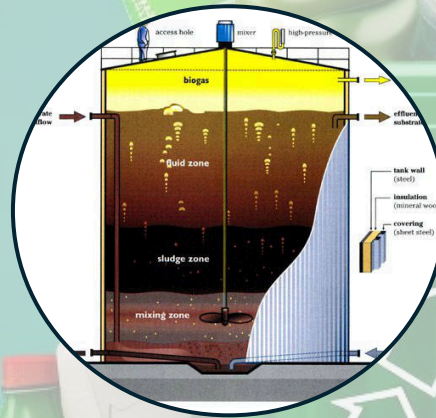
Can be used for
**Cooking &
electricity gen**

CRUCIAL ROLE IN **ENERGY CRISIS**

BIO-FUEL ACROSS THE WORLD



Biodegradable Plastics:
Certain plastics can be processed in biogas plants to produce biogas without ecological risk.



Anaerobic Digestion:
Breaks down biodegradable plastics without oxygen, creating biogas.

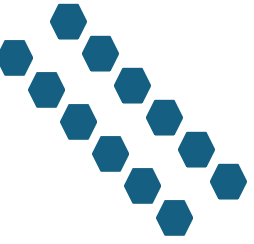


Biogas Digesters:
Can process organic waste, including certain plastics, to efficiently produce biogas.



Energy Recovery:
Contributes to waste management and renewable energy production for various uses.

ACTION PLAN



Collect Plastics
from Various
Sources

PROCUREMENT



Storage in
Central
Facilities



Undergo
Gasification
Procedure

PRODUCTION



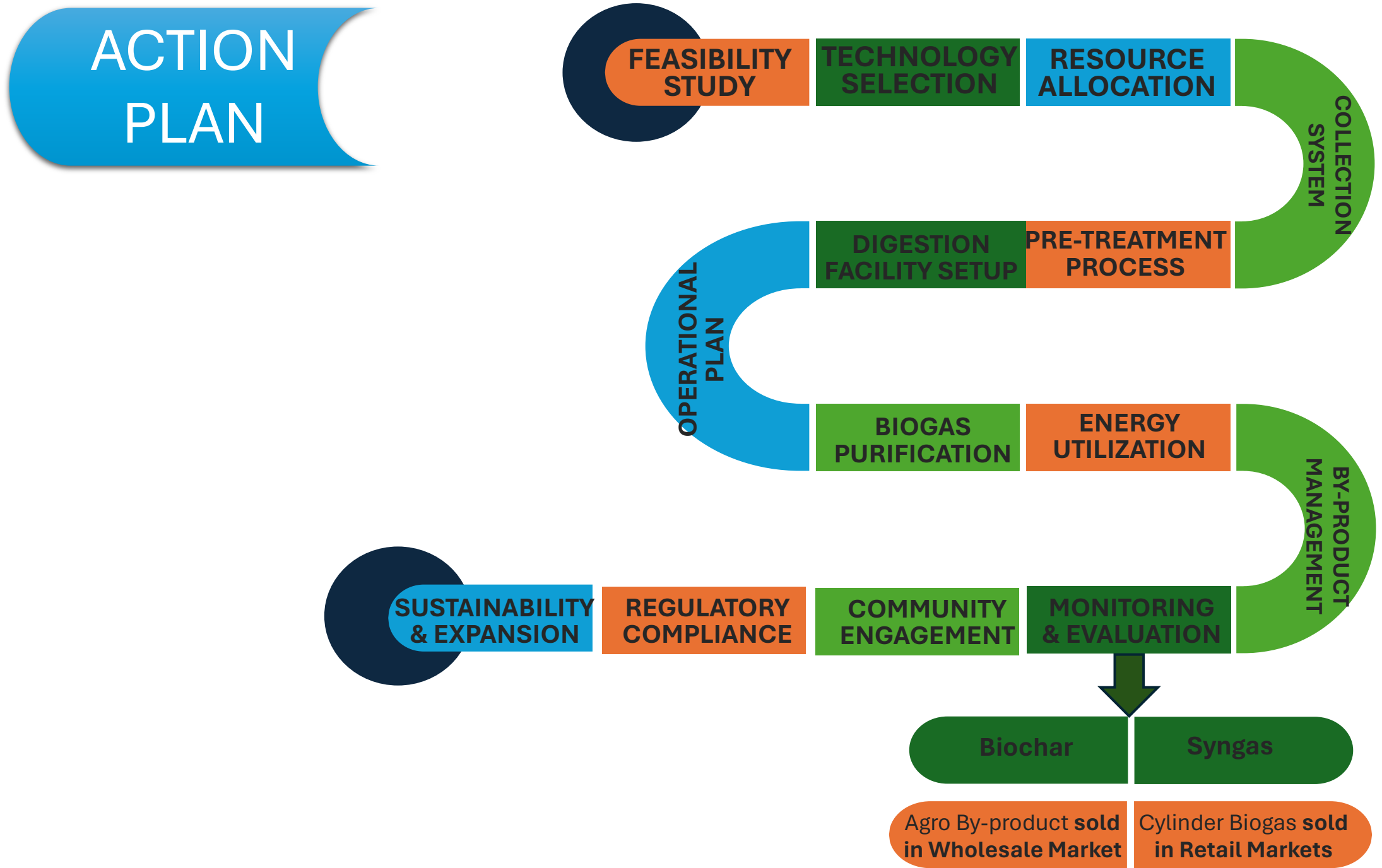
Package in Gas
Cylinders



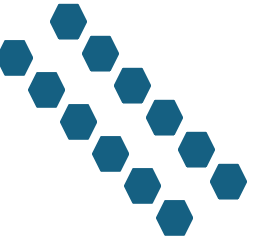
Distribute via
Distributors,
Dealers and E-
Commerce

DISTRIBUTION

PRODUCTION PROCESS



MARKETING AND CAMPAIGN ACTIVITY



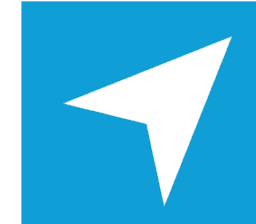
**Gas Cylinder
Dealers**

18,000 dealer
network within
Dhaka Metro City



E-Commerce

Sheba.xyz
Gasmonkey
Cylinderwala



**Direct to
Consumer**

Authorised
Distributors

Distribution Point



**Digital
Marketing**

Website and SEO
Social Media: LinkedIn, Twitter,
Facebook, and Instagram



**Partnership with
industry leaders**

To demonstrate the feasibility and
scalability of **plastic-to-energy**
solutions, showcasing successful
case studies and partnerships

MARKETING AND CAMPAIGN ACTIVITY

PROCUREMENT

Initial Partners are businesses that produce high waste / unit

- RESTAURANTS
- HOTELS
- OFFICES
- HOSTELS
- WEDDING HALLS
- WET MARKETS
- RECYCLING PARTNERS

Pipeline Expanding into Sourcing from Direct Consumers like Households



The Future of Plastics is Here

Danimer Scientific is making the future of plastics possible, from compostable polymers for today's sustainable products to achieving 100% biodegradable bioplastics for tomorrow.

[REQUEST MORE INFO](#)

SCROLL

MARKETING AND CAMPAIGN ACTIVITY



ESG IMPACT



ENVIRONMENT

Fossil Fuel
Dependence
Reduced

Organic Waste
Management



SOCIAL

Awareness for
Renewable
Energy

Contributes
sustainable city
development

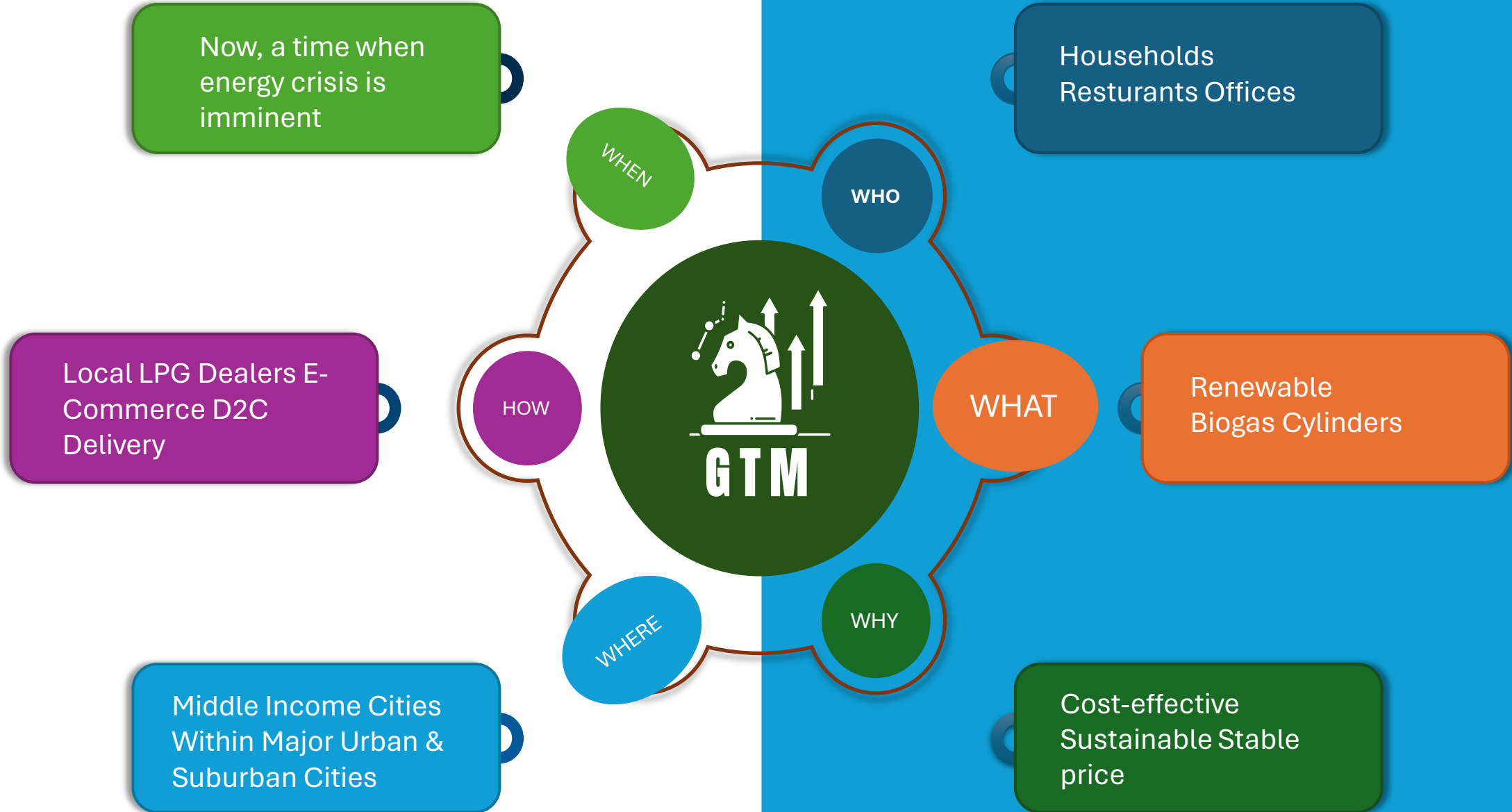


GOVERNMENT

Energy Import
Dependence
Reduced

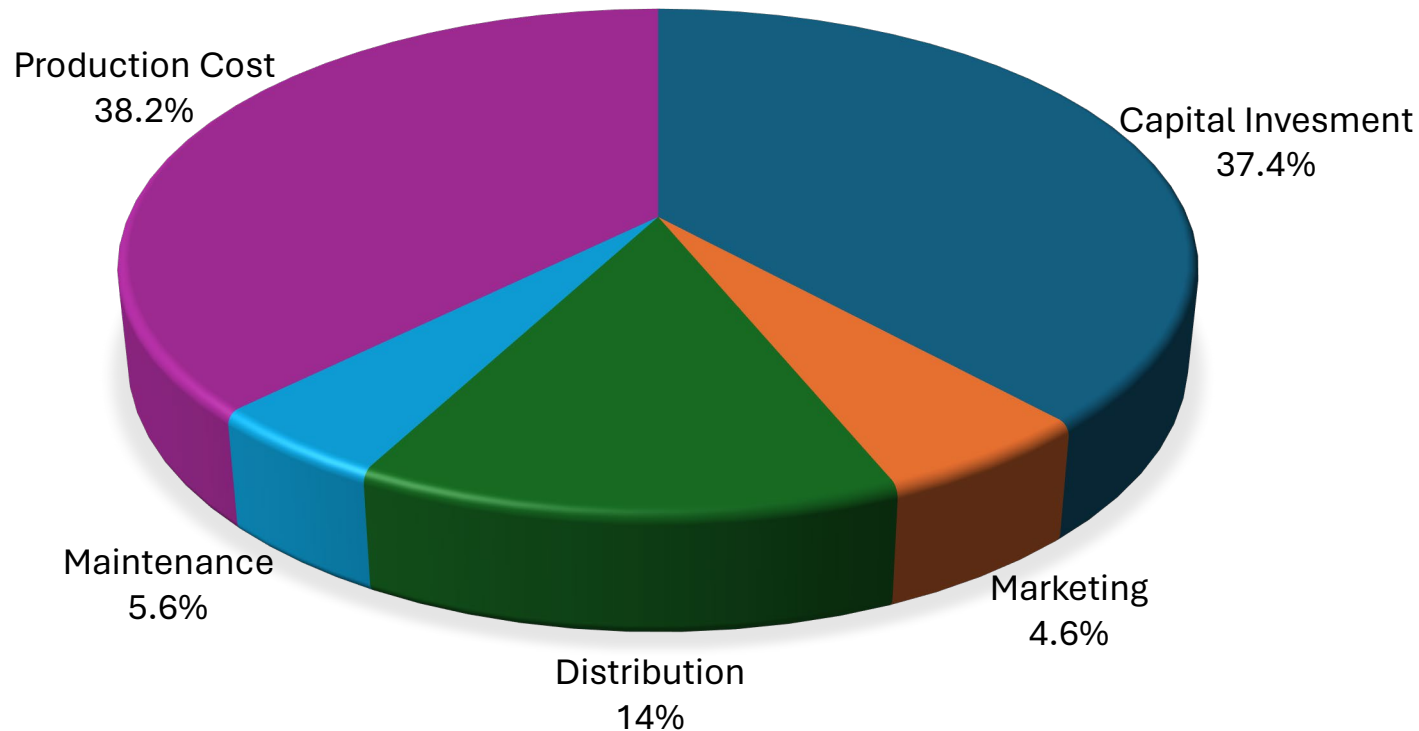
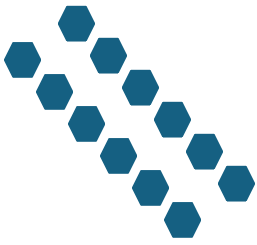
Proof of Concept
Renewable
Sources

MARKETING AND CAMPAIGN ACTIVITY



FINANCE

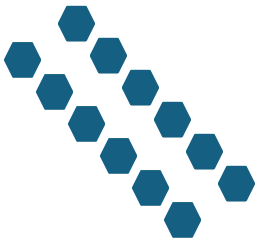
WE ARE RAISING
US \$1 MILLION



KEY MILESTONES

- CAPITAL INVESTMENT**
Facility 1 in year 1
Facility 2 in year 5
- DISTRIBUTION**
Expand outside Dhaka by year 3
- PRODUCTION**
Acquire 5% of the Domestic Cooking Gas Market
- MARKETING & SALES**
Sell 115k+ units (1.4M ltrs) by year 1

FINANCE



TOTAL AVAILABLE MARKET

\$134.76 BILLION

Global biomass power market size est. 2022 source, precedence Research

SERVICEABLE AVAILABLE MARKET

\$516 MILLION

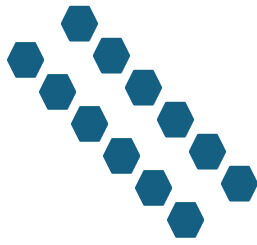
4.3 Million LPG consumers in BD x Average \$10 monthly expense per consumer x 12 month.

SERVICEABLE OBTAINABLE MARKET

~\$50 MILLION

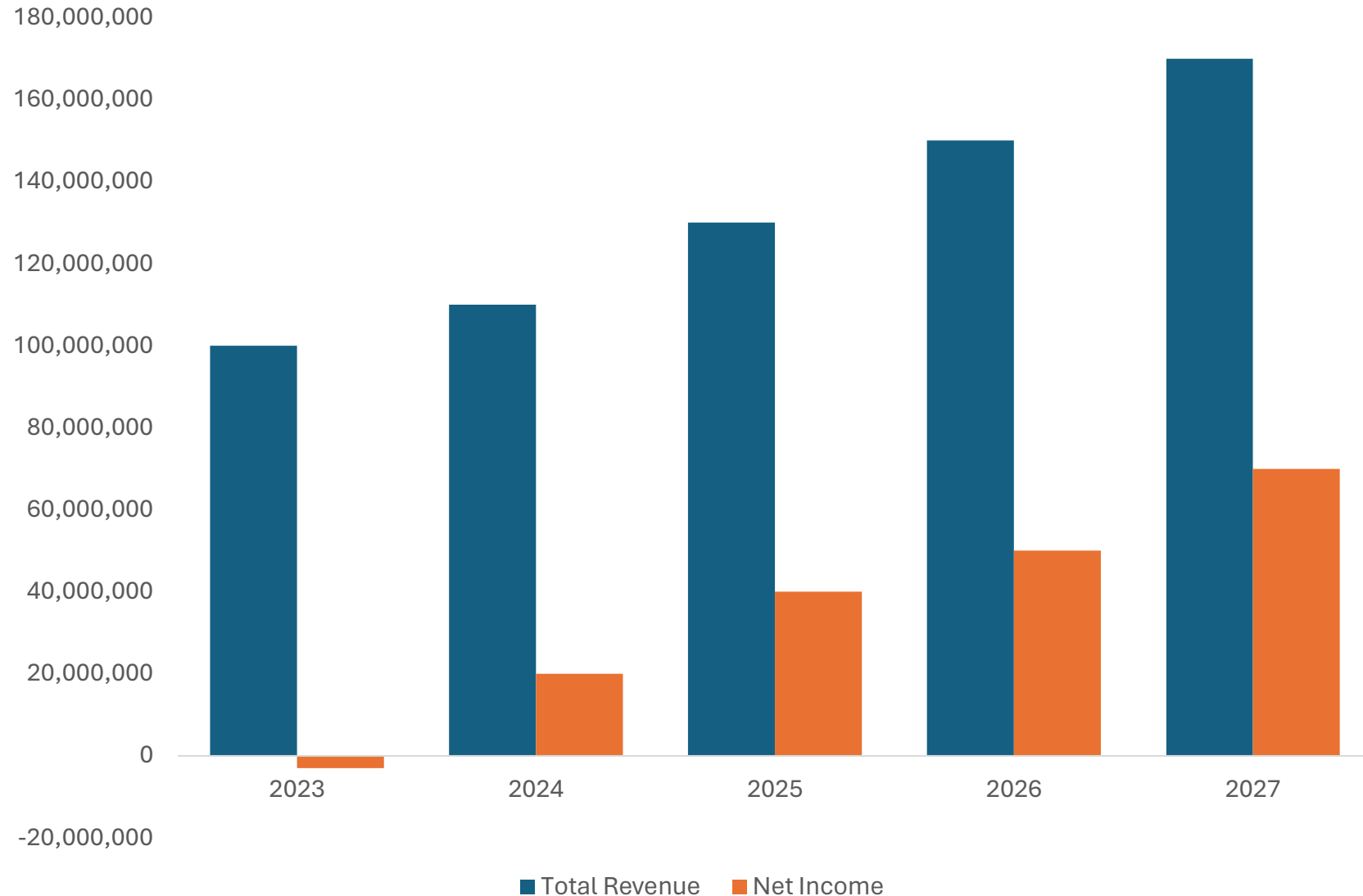
10% of SAM= \$51.6 Million

FINANCE

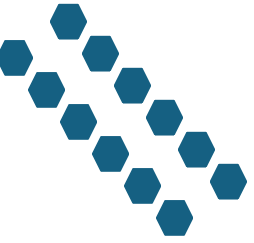


KEY INSIGHTS

- Payback Period**
3.96 years
- Assumed Growth**
Y1-3: 10% YoY
Y4-5: 15% YoY
- Price Increment**
Y3: Syngas (70->75)
Y4: Biochar (15->18)
- IRR 5Y**
12.38%



FEASIBILITY



BIO-GAS CYLINDERS

70 BDT/L at 56% Margin



POLYCYCLE POWER

BDT 850

Small to Medium Households

POLYCYCLE POWER

BDT 2,500

Medium to Large Households

POLYCYCLE POWER

BDT 3,200

Restaurants/Hotels & Commercial Users

29% cheaper than LPG (1,176 BDT/ 12 LTR)



BIO-CHAR

By-Product of
Gasification Process

BDT 15/KG



Alternative revenue stream



Sold wholesale to farmers as
fertilizer



Create a 'green' process in
production



FEASIBILITY

END PRODUCT COMPARISON

To Alternative LPG Cylinder

Biogas is a more **Cost-effective renewable** for LPG gas for specific use-cases such as Cooking & powering Electricity Generators

- PRICE
- CARBON FOOTPRINT
- EFFICIENCY
- USAGE POTENTIAL
- RENEWABLE SOURE

BIOGAS
LOW & STABLE
LOW
HIGH
FOCUSED
YES

- HIGH & VOLATILE
- LOW
- HIGH
- VERSATILE
- NO



LPG



THANK YOU

